

Question:

A 35-year-old African American man presents with fever, abdominal pain, and severe weakness since yesterday. On physical examination, the patient is jaundiced and shows a generalized pallor. Past medical history is significant for recently receiving anti-malaria prophylaxis before visiting Nigeria. Laboratory tests show normal glucose-6-phosphate dehydrogenase (G6PD) levels. Peripheral smear shows the presence of bite cells and Heinz bodies. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

- A. Autoimmune hemolytic anemia
- B. Microangiopathic hemolytic anemia
- C. Paroxysmal nocturnal hemoglobinuria (PNH)
- D. Glucose-6-phosphate-dehydrogenase (G6PD) deficiency

Reasoning + Answer:

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The patient is a 35-year-old African American man with fever, abdominal pain, weakness, jaundice, generalized pallor, and laboratory findings of bite cells and Heinz bodies on peripheral smear. He recently received anti-malaria prophylaxis before traveling to Nigeria, which is a key clue.

G6PD deficiency is a common cause of hemolytic anemia, especially in young males and those in malaria-endemic regions. The typical presentation includes hemolytic anemia triggered by certain drugs, infections, or foods. The presence of bite cells and Heinz bodies is classic for G6PD deficiency — bite cells result from intravascular hemolysis, and Heinz bodies are denatured hemoglobin precipitates seen in oxidative hemolysis.

Although the question states that G6PD levels are "normal," this is likely a trick — the test may have been performed in the context of the cell's function (as G6PD activity is more relevant than enzyme levels), and the clinical presentation (after anti-malaria prophylaxis, with hemolysis and characteristic peripheral smear findings) strongly suggests a deficiency.

Moreover, the patient's exposure to anti-malarial drugs (such as primaquine) is a well-known trigger of hemolytic anemia in those with G6PD deficiency.

Autoimmune hemolytic anemia (A) is less likely without evidence of autoantibodies or a positive direct Coombs test. Microangiopathic hemolytic anemia (B) often presents with schistocytes, not bite cells or Heinz bodies. PNH (C) involves complement dysregulation and is associated with schistocytes and low platelets, not typical of this presentation.

Therefore, despite the stated "normal G6PD levels," the clinical context — exposure to anti-malarial drugs, classic lab findings (bite cells, Heinz bodies), and ethnicity — makes G6PD deficiency the most likely diagnosis.

< answer > Glucose - 6 -phosphate-dehydrogenase (G 6 PD) deficiency </answer>

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